2.8 Function IO Description

Table 2-3 Function IO description

Pin	Pin Name	Func0	Func1	Func2	Func3	Func4	Func5	Func6	Func7	Func8	Pad Type①	Def③	Pull	Drive Strength②	INT	Power Domain
121	OSC_XIN	OSC_XIN									I	I	N/A	N/A		PLL
122	OSC_XOUT	OSC_XOUT									O	O	N/A	N/A		
119	OSC_CLK_OUT/REF_CLK0_OUT/GPIO0_D0_d	GPIO0_D0	OSC_CLK_OUT	REF_CLK0_OUT							IO	I	down	Level1	√	
10	NPOR	NPOR									I	I	N/A	N/A		PMUIO_VCC3V3
21	SAI0_LRCK/RM_IO0/GPIO0_A0_u	GPIO0_A0	SAI0_LRCK						RM_IO0		IO	I	up	Level2	√	
20	SAI0_SCLK/RM_IO1/GPIO0_A1_d	GPIO0_A1	SAI0_SCLK						RM_IO1		IO	I	down	Level2	√	
19	SAI0_MCLK/RM_IO2/GPIO0_A2_u	GPIO0_A2	SAI0_MCLK						RM_IO2		IO	I	up	Level2	√	
18	SAI0_SDO/RM_IO3/GPIO0_A3_d	GPIO0_A3	SAI0_SDO						RM_IO3		IO	I	down	Level2	√	
17	SAI0_SDI0/RM_IO4/GPIO0_A4_d	GPIO0_A4	SAI0_SDI0						RM_IO4		IO	I	down	Level2	√	
16	SAI0_SDI1/RM_IO5/GPIO0_A5_d	GPIO0_A5	SAI0_SDI1						RM_IO5		IO	I	down	Level2	√	
15	SAI0_SDI2/RM_IO6/GPIO0_A6_d	GPIO0_A6	SAI0_SDI2						RM_IO6		IO	I	down	Level2	√	
14	SAI0_SDI3/SPI1_CSN1/RM_IO7/GPIO0_A7_d	GPIO0_A7	SAI0_SDI3	SPI1_CSN1					RM_IO7		IO	I	down	Level2	√	
13	SAI1_MCLK/SPI1_CLK/RM_IO8/GPIO0_B0_d	GPIO0_B0	SAI1_MCLK	SPI1_CLK					RM_IO8		IO	I	down	Level2	√	
12	SAI1_SCLK/SPI1_MOSI/RM_IO9/GPIO0_B1_d	GPIO0_B1	SAI1_SCLK	SPI1_MOSI					RM_IO9		IO	I	down	Level2	√	
8	SAI1_LRCK/SPI1_MISO/RM_IO10/GPIO0_B2_d	GPIO0_B2	SAI1_LRCK	SPI1_MISO					RM_IO10		IO	I	down	Level2	√	
7	SAI1_SDI/RM_IO11/GPIO0_B3_d	GPIO0_B3	SAI1_SDI						RM_IO11		IO	I	down	Level2	√	
6	SAI1_SDO0/RM_IO12/GPIO0_B4_d	GPIO0_B4	SAI1_SDO0						RM_IO12		IO	I	down	Level2	√	
5	SAI1_SDO1/RM_IO13/GPIO0_B5_d	GPIO0_B5	SAI1_SDO1						RM_IO13		IO	I	down	Level2	√	
4	SAI1_SDO2/SPI1_CSN0/RM_IO14/GPIO0_B6_d	GPIO0_B6	SAI1_SDO2	SPI1_CSN0					RM_IO14		IO	I	down	Level2	√	
3	SAI1_SDO3/SPI0_CSN1/RM_IO15/GPIO0_B7_d	GPIO0_B7	SAI1_SDO3	SPI0_CSN1					RM_IO15		IO	I	down	Level2	√	
2	SPI0_CLK/RM_IO16/GPIO0_C0_d	GPIO0_C0		SPI0_CLK					RM_IO16		IO	I	down	Level2	√	
1	SPI0_MOSI/RM_IO17/GPIO0_C1_d	GPIO0_C1		SPI0_MOSI					RM_IO17		IO	I	down	Level2	√	
128	REF_CLK1_OUT/SPI0_MISO/RM_IO18/GPIO0_C2_d	GPIO0_C2	REF_CLK1_OUT	SPI0_MISO					RM_IO18		IO	I	down	Level2	√	
127	ETH_CLK1_25M_OUT/SPI0_CSN0/RM_IO19/GPIO0_C3_d	GPIO0_C3	ETH_CLK1_25M_OUT	SPI0_CSN0					RM_IO19		IO	I	down	Level2	√	
126	ETH_CLK0_25M_OUT/AUPLL_CLK_IN/RM_IO20/GPIO0_C4_d	GPIO0_C4	ETH_CLK0_25M_OUT	AUPLL_CLK_IN					RM_IO20		IO	I	down	Level2	√	
125	CPU_AVS/RM_IO21/GPIO0_C5_z	GPIO0_C5	CPU_AVS						RM_IO21		IO	I	z	Level2	√	
124	UART0_TX/JTAG_TCK_M1/RM_IO22/GPIO0_C6_u	GPIO0_C6	UART0_TX	JTAG_TCK_M1					RM_IO22		IO	I	up	Level2	√	
123	UART0_RX/JTAG_TMS_M1/RM_IO23/GPIO0_C7_u	GPIO0_C7	UART0_RX	JTAG_TMS_M1					RM_IO23		IO	I	up	Level2	√	
83	VO_LCDC_DEN/DSMC_CLKP/FLEXBUS1_D0/GPIO1_A0_d	GPIO1_A0	VO_LCDC_DEN	DSMC_CLKP	FLEXBUS1_D0						IO	I	down	Level2	√	VCCIO1_VCC
84	VO_LCDC_VSYNC/DSMC_CLKN/FLEXBUS1_D1/DSMC_INT0/DSMC_SLV_INT/GPIO1_A1_d	GPIO1_A1	VO_LCDC_VSYNC	DSMC_CLKN	FLEXBUS1_D1	DSMC_INT0				DSMC_SLV_INT	IO	I	down	Level2	√	
88	VO_LCDC_HSYNC/DSMC_DQS0/FLEXBUS1_D2/GPIO1_A2_d	GPIO1_A2	VO_LCDC_HSYNC	DSMC_DQS0	FLEXBUS1_D2						IO	I	down	Level2	√	
89	VO_LCDC_CLK/DSMC_D0/FLEXBUS1_D3/GPIO1_A3_d	GPIO1_A3	VO_LCDC_CLK	DSMC_D0	FLEXBUS1_D3						IO	I	down	Level2	√	
90	VO_LCDC_D23/DSMC_D1/FLEXBUS1_D4/GPIO1_A4_d	GPIO1_A4	VO_LCDC_D23	DSMC_D1	FLEXBUS1_D4						IO	I	down	Level2	√	
91	VO_LCDC_D22/DSMC_D2/FLEXBUS1_D5/GPIO1_A5_d	GPIO1_A5	VO_LCDC_D22	DSMC_D2	FLEXBUS1_D5						IO	I	down	Level2	√	
93	VO_LCDC_D21/DSMC_D3/FLEXBUS1_D6/GPIO1_A6_d	GPIO1_A6	VO_LCDC_D21	DSMC_D3	FLEXBUS1_D6						IO	I	down	Level2	√	
94	VO_LCDC_D20/DSMC_D4/FLEXBUS1_D7/GPIO1_A7_d	GPIO1_A7	VO_LCDC_D20	DSMC_D4	FLEXBUS1_D7						IO	I	down	Level2	√	
95	VO_LCDC_D19/DSMC_D5/FLEXBUS1_D8/FLEXBUS0_CSN_M0/GPIO1_B0_d	GPIO1_B0	VO_LCDC_D19	DSMC_D5	FLEXBUS1_D8		FLEXBUS0_CSN_M0				IO	I	down	Level2	√	
96	VO_LCDC_D18/DSMC_CSN1/FLEXBUS1_D9/FLEXBUS1_CSN_M0/UART5_CTSN_M1/RM_IO24/GPIO1_B1_d	GPIO1_B1	VO_LCDC_D18	DSMC_CSN1	FLEXBUS1_D9		FLEXBUS1_CSN_M0	UART5_CTSN_M1	RM_IO24		IO	I	down	Level2	√	
97	VO_LCDC_D17/DSMC_INT2/FLEXBUS1_D10/FLEXBUS0_D15/FLEXBUS0_CSN_M1/SAI2_SCLK_M1/RM_IO25/GPIO1_B2_d	GPIO1_B2	VO_LCDC_D17	DSMC_INT2	FLEXBUS1_D10	FLEXBUS0_D15	FLEXBUS0_CSN_M1	SAI2_SCLK_M1	RM_IO25		IO	I	down	Level2	√	

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Pin	Pin Name	Func0	Func1	Func2	Func3	Func4	Func5	Func6	Func7	Func8	Pad Type①	Def③	Pull	Drive Strength②	INT	Power Domain
98	VO_LCDC_D16/DSMC_INT3/FLEXBUS1_D11/FLEXBUS0_D14/FLEXBUS1_CSN_M1/SAI2_LRCK_M1/RM_IO26/GPIO1_B3_d	GPIO1_B3	VO_LCDC_D16	DSMC_INT3	FLEXBUS1_D11	FLEXBUS0_D14	FLEXBUS1_CSN_M1	SAI2_LRCK_M1	RM_IO26		IO	I	down	Level2	√	
99	VO_LCDC_D15/DSMC_D6/FLEXBUS1_D12/FLEXBUS0_D13/FLEXBUS0_CSN_M2/GPIO1_B4_d	GPIO1_B4	VO_LCDC_D15	DSMC_D6	FLEXBUS1_D12	FLEXBUS0_D13	FLEXBUS0_CSN_M2				IO	I	down	Level2	√	
100	VO_LCDC_D14/DSMC_D7/FLEXBUS1_D13/FLEXBUS0_D12/FLEXBUS1_CSN_M2/GPIO1_B5_d	GPIO1_B5	VO_LCDC_D14	DSMC_D7	FLEXBUS1_D13	FLEXBUS0_D12	FLEXBUS1_CSN_M2				IO	I	down	Level2	√	
101	VO_LCDC_D13/DSMC_CSN0/FLEXBUS1_D14/FLEXBUS0_D11/FLEXBUS0_CSN_M3/GPIO1_B6_d	GPIO1_B6	VO_LCDC_D13	DSMC_CSN0	FLEXBUS1_D14	FLEXBUS0_D11	FLEXBUS0_CSN_M3				IO	I	down	Level2	√	
102	VO_LCDC_D12/DSMC_RDYN/FLEXBUS1_D15/FLEXBUS0_D10/FLEXBUS1_CSN_M3/GPIO1_B7_d	GPIO1_B7	VO_LCDC_D12	DSMC_RDYN	FLEXBUS1_D15	FLEXBUS0_D10	FLEXBUS1_CSN_M3				IO	I	down	Level2	√	
104	VO_LCDC_D11/DSMC_RESETN/FLEXBUS1_CLK/DSMC_INT1/FLEXBUS0_CSN_M4/DSMC_SLV_CLK/GPIO1_C0_d	GPIO1_C0	VO_LCDC_D11	DSMC_RESETN	FLEXBUS1_CLK	DSMC_INT1	FLEXBUS0_CSN_M4			DSMC_SLV_CLK	IO	I	down	Level2	√	
105	VO_LCDC_D10/DSMC_D8/FLEXBUS0_CLK/DSM_AUD_RN_M0/FLEXBUS1_CSN_M4/SAI2_MCLK_M1/DSMC_SLV_DQS0/GPIO1_C1_d	GPIO1_C1	VO_LCDC_D10	DSMC_D8	FLEXBUS0_CLK	DSM_AUD_RN_M0	FLEXBUS1_CSN_M4	SAI2_MCLK_M1		DSMC_SLV_DQS0	IO	I	down	Level2	√	
106	VO_LCDC_D9/DSMC_D9/FLEXBUS0_D9/DSM_AUD_RP_M0/FLEXBUS0_CSN_M5/SAI2_SDI_M1/RM_IO27/DSMC_SLV_D0/GPIO1_C2_d	GPIO1_C2	VO_LCDC_D9	DSMC_D9	FLEXBUS0_D9	DSM_AUD_RP_M0	FLEXBUS0_CSN_M5	SAI2_SDI_M1	RM_IO27	DSMC_SLV_D0	IO	I	down	Level2	√	
107	VO_LCDC_D8/DSMC_D10/FLEXBUS0_D8/FLEXBUS1_CSN_M5/SAI2_SDO_M1/RM_IO28/DSMC_SLV_D1/GPIO1_C3_d	GPIO1_C3	VO_LCDC_D8	DSMC_D10	FLEXBUS0_D8		FLEXBUS1_CSN_M5	SAI2_SDO_M1	RM_IO28	DSMC_SLV_D1	IO	I	down	Level2	√	
108	VO_LCDC_D7/DSMC_D11/FLEXBUS0_D7/DSMC_SLV_D2/GPIO1_C4_d	GPIO1_C4	VO_LCDC_D7	DSMC_D11	FLEXBUS0_D7					DSMC_SLV_D2	IO	I	down	Level2	√	
109	VO_LCDC_D6/DSMC_D12/FLEXBUS0_D6/DSMC_SLV_D3/GPIO1_C5_d	GPIO1_C5	VO_LCDC_D6	DSMC_D12	FLEXBUS0_D6					DSMC_SLV_D3	IO	I	down	Level2	√	
113	VO_LCDC_D5/DSMC_D13/FLEXBUS0_D5/DSMC_SLV_D4/GPIO1_C6_d	GPIO1_C6	VO_LCDC_D5	DSMC_D13	FLEXBUS0_D5					DSMC_SLV_D4	IO	I	down	Level2	√	
114	VO_LCDC_D4/DSMC_D14/FLEXBUS0_D4/DSMC_SLV_D5/GPIO1_C7_d	GPIO1_C7	VO_LCDC_D4	DSMC_D14	FLEXBUS0_D4					DSMC_SLV_D5	IO	I	down	Level2	√	
115	VO_LCDC_D3/DSMC_D15/FLEXBUS0_D3/DSM_AUD_LN_M0/DSMC_SLV_D6/GPIO1_D0_d	GPIO1_D0	VO_LCDC_D3	DSMC_D15	FLEXBUS0_D3	DSM_AUD_LN_M0				DSMC_SLV_D6	IO	I	down	Level2	√	
116	VO_LCDC_D2/DSMC_DQS1/FLEXBUS0_D2/DSM_AUD_LP_M0/UART5_RTSN_M1/RM_IO29/DSMC_SLV_D7/GPIO1_D1_d	GPIO1_D1	VO_LCDC_D2	DSMC_DQS1	FLEXBUS0_D2	DSM_AUD_LP_M0		UART5_RTSN_M1	RM_IO29	DSMC_SLV_D7	IO	I	down	Level2	√	
117	VO_LCDC_D1/DSMC_CSN2/FLEXBUS0_D1/UART5_TX_M1/RM_IO30/DSMC_SLV_CSN0/GPIO1_D2_d	GPIO1_D2	VO_LCDC_D1	DSMC_CSN2	FLEXBUS0_D1			UART5_TX_M1	RM_IO30	DSMC_SLV_CSN0	IO	I	down	Level2	√	
118	VO_LCDC_D0/DSMC_CSN3/FLEXBUS0_D0/UART5_RX_M1/RM_IO31/DSMC_SLV_RDYN/GPIO1_D3_d	GPIO1_D3	VO_LCDC_D0	DSMC_CSN3	FLEXBUS0_D0			UART5_RX_M1	RM_IO31	DSMC_SLV_RDYN	IO	I	down	Level2	√	
59	FSPI_CSN/GPIO2_A0_u	GPIO2_A0	FSPI_CSN								IO	I	up	Level2	√	VCCIO2_VCC
54	FSPI_CLK/GPIO2_A1_d	GPIO2_A1	FSPI_CLK								IO	I	down	Level3	√	
55	FSPI_D0/GPIO2_A2_u	GPIO2_A2	FSPI_D0								IO	I	up	Level2	√	
58	FSPI_D1/GPIO2_A3_u	GPIO2_A3	FSPI_D1								IO	I	up	Level2	√	
57	FSPI_D2/GPIO2_A4_u	GPIO2_A4	FSPI_D2								IO	I	up	Level2	√	
53	FSPI_D3/GPIO2_A5_u	GPIO2_A5	FSPI_D3								IO	I	up	Level2	√	
50	ETH_RMII0_RXD0/SPI2_CLK/GPIO2_B0_d	GPIO2_B0	ETH_RMII0_RXD0	SPI2_CLK							IO	I	down	Level2	√	VCCIO3_VCC
49	ETH_RMII0_RXD1/SPI2_CSN/GPIO2_B1_d	GPIO2_B1	ETH_RMII0_RXD1	SPI2_CSN							IO	I	down	Level2	√	
48	ETH_RMII0_CLK/SPI2_MOSI/GPIO2_B2_d	GPIO2_B2	ETH_RMII0_CLK	SPI2_MOSI							IO	I	down	Level2	√	
47	ETH_RMII0_TXD0/SPI2_MISO/GPIO2_B3_d	GPIO2_B3	ETH_RMII0_TXD0	SPI2_MISO							IO	I	down	Level2	√	
46	ETH_RMII0_TXD1/DSM_AUD_RN_M1/SAI3_SCLK/GPIO2_B4_d	GPIO2_B4	ETH_RMII0_TXD1	DSM_AUD_RN_M1	SAI3_SCLK						IO	I	down	Level2	√	
45	ETH_RMII0_TXEN/DSM_AUD_RP_M1/SAI3_LRCK/GPIO2_B5_d	GPIO2_B5	ETH_RMII0_TXEN	DSM_AUD_RP_M1	SAI3_LRCK						IO	I	down	Level2	√	
44	ETH_RMII0_MDC/DSM_AUD_LN_M1/SAI3_SDI/GPIO2_B6_d	GPIO2_B6	ETH_RMII0_MDC	DSM_AUD_LN_M1	SAI3_SDI						IO	I	down	Level2	√	
43	ETH_RMII0_MDIO/DSM_AUD_LP_M1/SAI3_SDO/GPIO2_B7_d	GPIO2_B7	ETH_RMII0_MDIO	DSM_AUD_LP_M1	SAI3_SDO						IO	I	down	Level2	√	
42	ETH_RMII0_RXDVCRS/SAI3_MCLK/GPIO2_C0_d	GPIO2_C0	ETH_RMII0_RXDVCRS		SAI3_MCLK						IO	I	down	Level2	√	
37	SDMMC_CLK/GPIO3_A0_d	GPIO3_A0	SDMMC_CLK								IO	I	down	Level3	√	VCCIO4_VCC
38	SDMMC_CMD/GPIO3_A1_d	GPIO3_A1	SDMMC_CMD								IO	I	down	Level2	√	
36	SDMMC_D0/GPIO3_A2_d	GPIO3_A2	SDMMC_D0								IO	I	down	Level2	√	
35	SDMMC_D1/TEST_CLK_OUT/GPIO3_A3_d	GPIO3_A3	SDMMC_D1	TEST_CLK_OUT							IO	I	down	Level2	√	
41	SDMMC_D2/JTAG_TCK_M0/GPIO3_A4_d	GPIO3_A4	SDMMC_D2	JTAG_TCK_M0							IO	I	down	Level2	√	

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Pin	Pin Name	Func0	Func1	Func2	Func3	Func4	Func5	Func6	Func7	Func8	Pad Type①	Def③	Pull	Drive Strength②	INT	Power Domain
40	SDMMC_D3/JTAG_TMS_M0/GPIO3_A5_d	GPIO3_A5	SDMMC_D3	JTAG_TMS_M0							IO	I	down	Level2	√	
34	SAI2_SDI_M0/ETH_RMII1_RXD0/GPIO3_A6_d	GPIO3_A6	SAI2_SDI_M0	ETH_RMII1_RXD0							IO	I	down	Level2	√	
33	SAI2_SCLK_M0/ETH_RMII1_RXD1/GPIO3_A7_d	GPIO3_A7	SAI2_SCLK_M0	ETH_RMII1_RXD1							IO	I	down	Level2	√	
32	SAI2_SDO_M0/ETH_RMII1_CLK/GPIO3_B0_d	GPIO3_B0	SAI2_SDO_M0	ETH_RMII1_CLK							IO	I	down	Level2	√	
31	SAI2_LRCK_M0/ETH_RMII1_TXD0/GPIO3_B1_d	GPIO3_B1	SAI2_LRCK_M0	ETH_RMII1_TXD0							IO	I	down	Level2	√	
30	UART5_CTSN_M0/ETH_RMII1_TXD1/GPIO3_B2_d	GPIO3_B2	UART5_CTSN_M0	ETH_RMII1_TXD1							IO	I	down	Level2	√	
29	UART5_RX_M0/ETH_RMII1_TXEN/GPIO3_B3_d	GPIO3_B3	UART5_RX_M0	ETH_RMII1_TXEN							IO	I	down	Level2	√	
28	UART5_TX_M0/ETH_RMII1_MDC/GPIO3_B4_d	GPIO3_B4	UART5_TX_M0	ETH_RMII1_MDC							IO	I	down	Level2	√	
27	UART5_RTSN_M0/ETH_RMII1_MDIO/GPIO3_B5_d	GPIO3_B5	UART5_RTSN_M0	ETH_RMII1_MDIO							IO	I	down	Level2	√	
26	SAI2_MCLK_M0/ETH_RMII1_RXDVCRS/GPIO3_B6_d	GPIO3_B6	SAI2_MCLK_M0	ETH_RMII1_RXDVCRS							IO	I	down	Level2	√	
81	MIPI_DPHY_DSI_TX_D0N/GPO4_A0_z	GPO4_A0	MIPI_DPHY_DSI_TX_D0N								O	O	Z			
82	MIPI_DPHY_DSI_TX_D0P/GPO4_A1_z	GPO4_A1	MIPI_DPHY_DSI_TX_D0P								O	O	Z			
79	MIPI_DPHY_DSI_TX_D1N/GPO4_A2_z	GPO4_A2	MIPI_DPHY_DSI_TX_D1N								O	O	Z			
80	MIPI_DPHY_DSI_TX_D1P/GPO4_A3_z	GPO4_A3	MIPI_DPHY_DSI_TX_D1P								O	O	Z			MIPI
77	MIPI_DPHY_DSI_TX_CLKN/GPO4_A4_z	GPO4_A4	MIPI_DPHY_DSI_TX_CLKN								O	O	Z			
78	MIPI_DPHY_DSI_TX_CLKP/GPO4_A5_z	GPO4_A5	MIPI_DPHY_DSI_TX_CLKP								O	O	Z			
63	SARADC_IN0/GPIO4_B0_z	GPIO4_B0	SARADC_IN0								IO	I	z	Level1	√	
62	SARADC_IN1/GPIO4_B1_z	GPIO4_B1	SARADC_IN1								IO	I	z	Level1	√	SARADC
61	SARADC_IN2/GPIO4_B2_z	GPIO4_B2	SARADC_IN2								IO	I	z	Level1	√	
60	SARADC_IN3/GPIO4_B3_z	GPIO4_B3	SARADC_IN3								IO	I	z	Level1	√	
72	USB20_OTG0_DP	USB20_OTG0_DP									A					
71	USB20_OTG0_DM	USB20_OTG0_DM									A					
74	USB20_OTG1_DP	USB20_OTG1_DP									A					USB
73	USB20_OTG1_DM	USB20_OTG1_DM									A					
68	ACODEC_ADC_INP	ACODEC_ADC_INP									A					
67	ACODEC_ADC_INN	ACODEC_ADC_INN									A					
66	ACODEC_ADC_VCM	ACODEC_ADC_VCM									A					ACODEC ADC
65	ACODEC_ADC_AVDD1V6	ACODEC_ADC_AVDD1V6									A					

Notes:

□: Pad types: I = input, O = output, I/O = input/output (bidirectional)
AP = Analog Power, AG = Analog Ground
DP = Digital Power, DG = Digital Ground
A = Analog

□: Output Drive Unit is mA, only Digital IO has drive value;

□: Reset state: I = input, O = output;

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